# Piter Z. Garcia Bautista

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## Research Profile

Graduate researcher focused on machine learning for diagnostics and decision support under noisy, heterogeneous, and partially observed data. Background includes an M.S. in Computer Science, current M.S. work in Data Science, industry healthcare AI experience, and computational biology training. Current work emphasizes reproducibility, robust evaluation, fairness-aware analysis, and translational relevance to clinical settings.

## Research Interests

Machine learning for precision medicine and diagnostics; robust learning under missing or uncertain context; fairness-aware and trustworthy AI for high-stakes decision support; reproducible computational evaluation systems.

## Education

**Rochester Institute of Technology** — Expected Aug 2026
*Master of Science, Data Science* — *Rochester, NY*

- Relevant coursework: Bioinformatics Algorithms, Data-Driven Knowledge Discovery, Neural Networks, Applied Statistics, Software Engineering for Data Science.

**Rochester Institute of Technology** — 2015
*Master of Science, Computer Science* — *Rochester, NY*

- Concentrations: Artificial Intelligence, Big Data Analytics, Computer Graphics.

**University of Rochester, Warner School of Education** — In progress
*Graduate study, Teaching Computer Science K–12 (M.S. track)* — *Rochester, NY*

- NSF Robert Noyce Scholar; Advanced Certificate in Leadership in Disability and Inclusive Practices.

## Research and Professional Experience

**Rochester Institute of Technology, Software Engineering Department** — Fall 2025 – Present
*Graduate Assistant, Quantum and AI Research* — *Rochester, NY*

- Build reproducible machine learning evaluation pipelines across local, Colab, and GCP environments with structured logging and validation.
- Develop experimental bandit and decision-making systems for uncertain, heterogeneous settings, with emphasis on reliability and traceability.
- Produce manuscript-ready technical analyses and research artifacts for faculty-led projects.

**Rochester Institute of Technology** — 2026
*Computational Biology Research Projects (BIO614 and BIO550)* — *Rochester, NY*

- Led substantial computational work in BIO614, including formal manuscript development in the project *BIO614-FinalProjectProposal* (Overleaf), emphasizing algorithm implementation, structured evaluation, and scientific reporting.

- Complemented BIO614 with BIO550 RNA-seq and differential-expression analysis, including reproducible preprocessing, quality-control validation, and biological interpretation.

**VEDADATA Inc.** Sep 2022 – Aug 2024
*Data Solutions Engineer* *New York, NY*

- Designed scalable Python and pandas workflows for ingestion, validation, preprocessing, and analytics operations.
- Strengthened production data reliability through structured quality checks and statistical diagnostics.

**VIOME Inc.** Jan 2021 – Sep 2022
*AI Data Solutions Engineer* *New York, NY*

- Developed healthcare-oriented machine learning workflows for preprocessing, feature engineering, model experimentation, and evaluation.
- Supported reproducible analytics workflows in AWS-based environments for predictive health applications.

## Selected Technical Writing

- *Fairness-Aware Bandits for Network Routing in Quantum and Clinical Settings* (DSCI601 project proposal, manuscript-style report).
- *BIO614-FinalProjectProposal* (Overleaf) and companion BIO550 computational biology analyses on RNA structure and RNA-seq workflows, with reproducible diagnostics-oriented interpretation.

## Technical Competencies

**Programming:** Python, R, Java, C++, C#, SQL, MATLAB, JavaScript, Bash
**Machine Learning:** PyTorch, TensorFlow, scikit-learn, XGBoost, model evaluation, uncertainty-aware analysis, fairness-aware ML
**Data and Statistics:** pandas, NumPy, SciPy, statistical modeling, reproducible benchmarking
**Research Tooling:** Git/GitHub, Linux, Docker, LaTeX, AWS, GCP

## Selected Fit for KTH Single-Cell Position

- Combined computational biology evidence from BIO614 and BIO550, including manuscript-style technical writing and RNA-seq differential-expression analysis.
- Strong Python-based machine learning and reproducibility background for robust method development on high-dimensional, heterogeneous biomedical data.
- Clear research motivation for AI and machine learning methods that improve interpretability and reliability in single-cell cancer-data analysis.